ASSIUT MOTORSPORT TEAM PRESENTS

D4S

Drive for Safety

AI Safety Technology for Commercial Vehicles

FROM ACCIDENT DETECTION → TO ACCIDENT PREVENTION

POSITIONING

An AI safety layer designed for commercial vehicles and adapted to the Egyptian market.

About Us

Assiut Motorsport Team is specialized in designing and manufacturing racing cars. Students from various majors at Assiut University are team members, using professional **CAD software** and **simulation tools** to push engineering boundaries.

CAD Design

Professional-grade modeling

Simulation

CFD & structural analysis

Manufacturing

End-to-end prototyping

Now applying our **motorsport engineering expertise** to solve real-world road safety challenges.

MISSION

Enhance automotive knowledge in **Upper Egypt** and create a new ground for engineering students to apply real-world skills.

VISION

Keep pace with the **industrial revolution** by connecting learning to the labor market through hands-on innovation.

D4S — OUR VISION

To make Egyptian roads safer by turning every vehicle into a system that **actively prevents accidents** — not just records them.

D4S — OUR MISSION

We help companies reduce accidents using AI that **monitors drivers and predicts risks** before they become incidents.

The Problem

Commercial vehicle risk begins inside the cab — long before a crash is recorded.

DRIVER STATE

Fatigue & Drowsiness

Long-haul drivers operating under exhaustion.

DRIVER BEHAVIOR

Distraction & Risky Driving

Speed patterns, harsh braking, lane discipline.

VEHICLE & ENVIRONMENT

Road & Collision Risk

Distance, velocity and surrounding hazards.

90%

of road accidents caused by **human error**

0

local AI safety solutions identified in **Egypt today**

Up to 30%

of fleet **operational cost** — repairs, insurance, downtime, liability

The Gap — and The Shift

Traditional fleet solutions focus mainly on tracking and fleet management.

WHAT THE MARKET ALREADY ANSWERS

“Where is the vehicle?”

- GPS / location tracking
- Fleet management dashboards
- Trip history & basic behavior logs
- Reporting **after** the event

THE LAYER THAT IS STILL MISSING

“How is risk developing?”

+ Is the driver fatigued or distracted?
+ Is risk escalating right now?
+ What warning should fire first?
+ What can be done **before** a collision?

From Accident Detection → To Accident Prevention

THE D4S PHILOSOPHY

From Tracking | To Predicting | To Preventing

The value is no longer the record of what happened – it is the seconds gained before it happens.

The Solution

D4S is an AI safety layer for commercial vehicles — designed to be integrated into the vehicle, not bolted onto a dashboard.

01

Driver Monitoring

Fatigue, drowsiness, distraction and risky behavior – plus driver health monitoring.

02

Vehicle & Environment

Cameras, radar and LIDAR sensing distance, velocity and surrounding hazards.

03

Predictive Risk Analysis

An AI model fuses all inputs and outputs probability scores per risk category.

04

Warning & Intervention

Graduated escalation from notification to alert and vehicle-level response.

Not one feature — a combination: Driver State + Driver Behavior + Vehicle Environment + Predictive AI + Intervention

How D4S Works

All sensors stream body & environment data simultaneously into one AI pipeline.

01 — Sensing
Cameras, radar, LIDAR, IR & bio-sensors

02 — Data Ingestion
Simultaneous multi-stream input

03 — AI Analysis
NVIDIA AI model fuses all inputs, runs deep-learning inference

04 — Risk Detection
Probability scores per risk category

05 — Warning
Dashboard, audio & haptic alerts

06 — Critical Event
Highest probability risk category

07 — Intervention / Alert
GPS emergency alert to dispatch

STEP-BY-STEP ACTION ESCALATION

Gradual actions – never extreme unless extreme probability. The system responds proportionally to the confidence of the risk it detects.

PROBABILITY PREDICTION – OUTPUT CLASSES

- Normal
- Fatigue
- Emergency

SOURCE: D4S TECHNICAL ARCHITECTURE – AI PIPELINE & STEP-BY-STEP ACTION SYSTEM

The AI System

Inference pipeline, model outputs and the escalation logic that turns a score into an action.

STAGE 01 — INPUT VECTORS FUSED BY THE MODEL

Driver State
Head pose, eye & blink, HR, SpO2, skin temp, GSR

Driver Behavior
Speed patterns, braking intensity, lane discipline

Vehicle / Environment
Distance, velocity, lane & traffic sign, 3D map

STAGE 02 — INFERENCE CORE

NVIDIA AI Model

Runs on Jetson Nano, on-board the vehicle

Fuses all inputs, runs **deep-learning inference**, and outputs a **probability score per risk category** — a single defensible judgement rather than isolated sensor alarms.

STAGE 03 — PROBABILITY PREDICTION OUTPUT CLASSES

Normal
Baseline — score logged

Fatigue
Alertness degrading

Emergency
Critical event detected

Fatigue model: Head Pose Estimation (OpenCV / MediaPipe) → alarm if tilt >25° for >2s · **SafeScore:** daily risk score per driver · **Health:** wellness & alertness, not medical diagnosis

STAGE 04 — STEP-BY-STEP ACTION ESCALATION

L1 – Low Risk
Dashboard notification + data logging

L2 – Medium Risk
Audio alert to driver + push to fleet manager

L3 – High Risk
Strong audio + haptic + progressive speed reduction

L4 – Critical
Full AEB + vehicle slowdown + GPS emergency alert to dispatch

MATURITY — STATED HONESTLY

Current: monitoring, scoring, risk detection and alerting (L1–L2).

Prototype / concept: ADAS / AEB and vehicle intervention (L3–L4), not yet validated on a real vehicle.

Technical Architecture

Smart Safety Retrofit Kit — three sensing layers feeding one AI brain.

1 — ENVIRONMENTAL PERCEPTION

The Outer Shield

Optical Cameras — traffic sign & lane	Radar — distance & velocity, all-weather	LIDAR — 3D mapping, collision avoidance

2 — DRIVER MONITORING SYSTEM (DMS)

The Inner Guard

IR Camera 60FPS + 850nm — eye & blink, works in darkness	NVIDIA Jetson Nano — AI brain	Raspberry Pi — auxiliary logic

: Fatigue: Head Pose Estimation (OpenCV / MediaPipe) — alarm if tilt >25° for >2s

3 — CONTACTLESS BIO-SENSING

Health intelligence

MAX30102 — HR + SpO2	**MLX90614** — skin temp	**Grove-GSR** — stress	**MPU6050** — IMU

: mmWave Radar — HR & respiration through clothing · MLX90640 Thermal — body heat mapping · PPG — cameras based HR

ALL SENSOR DATA FLOWS TO

NVIDIA AI Model

Fuses all inputs, runs deep-learning inference, outputs probability scores per risk category

ADAS FEATURES

- **LKA** Lane Keeping with torque sensor
- **BSM** Blind Spot Monitoring via SRR radar
- **ACC** Stop-and-Go adaptive cruise control
- **AEB** Automatic Emergency Braking (fail-safe)

ADAS / AEB functions are prototype-stage and intended for delivery through licensed OEM integration.

SOURCE: D4S TECHNICAL ARCHITECTURE — SENSING LAYERS & AI PIPELINE

Hardware & Manufacturing

The physical kit — compute, sensing modules and how the unit is actually built.

01 — COMPUTE & CONTROL BOARDS

NVIDIA Jetson Nano
AI brain for deep learning – on-board inference

Raspberry Pi
Auxiliary logic & management

02 — SENSING MODULES

ENVIRONMENT & CABIN VISION

- Optical Cameras – traffic sign & lane
- Radar – distance & velocity, all-weather
- LIDAR – 3D mapping
- SRR radar – blind spot
- IR Camera 60FPS + 850nm – eye & blink, works in darkness
- Torque sensor – lane keeping

CONTACTLESS BIO-SENSING

- **MAX30102** – HR + SpO2 via light
- **MLX90614** – skin temp (infrared)
- **Grove-GSR** – skin conductivity (stress)
- **MPU6050** – IMU movement / stillness

› Advanced: mmWave Radar – HR & respiration through clothing › MLX90640 Thermal – body heat mapping › fPPG – camera-based HR (micro-skin colour changes)

03 — HOW THE UNIT IS BUILT

Built in-house by the Assiut Motorsport Team using the same discipline applied to racing cars:

- **CAD design** – professional-grade modeling
- **Simulation** – CFD & structural analysis
- **Manufacturing** – end-to-end prototyping

PROTOTYPE — SINGLE UNIT BUILD

49,500 EGP

Initial R&D / prototype investment

PER-VEHICLE PRODUCTION COST

42,000 EGP – hardware & components

8,000 EGP – integration & setup

SOURCE: D4S TECHNICAL ARCHITECTURE & FINANCIAL PLAN — SMART SAFETY RETROFIT KIT

Why D4S?

Not a single feature nobody else has – a combination almost nobody assembles for commercial vehicles in Egypt.

01
Driver State
Fatigue, drowsiness, alertness

+

02
Driver Behavior
Speed, braking, lane discipline

+

03
Vehicle Environment
Cameras, radar, LIDAR sensing

+

04
Predictive AI
Risk scoring before the event

+

05
Inter-vention
Escalation to action

= The D4S Safety Layer

FOCUS 01
Commercial vehicles – not a general fleet platform

FOCUS 02
Egyptian market – local engineering, customization & support

FOCUS 03
OEM-ready integration – not only aftermarket installation

Target Market

DEPLOYMENT PATH

D4S → Company → Integration into vehicle → Required testing / approval → Vehicle goes to market

Primary customers: **licensed automotive manufacturers and authorized vehicle integrators** serving the commercial-vehicle market. The OEM / integrator is the primary customer or channel partner; fleets are the end users / beneficiaries.

PRIORITY TARGETS IN EGYPT

General Motors Egypt
Strong fit because of its commercial-vehicle presence

GB Auto / Ghabbour
Commercial vehicles, trucks and buses — multi-brand ecosystem

SMG Engineering / Scania
Strong fit with the heavy-duty commercial-vehicle market

Mercedes-Benz Egypt / MCV ecosystem – subject to verification
Promising: current manufacturing / integration structure to be confirmed

SECONDARY TARGETS

Mansour Automotive · EIM · Alfa Ezz Elarab – depending on commercial-vehicle focus and connected-vehicle capability.

TARGET VEHICLE SEGMENTS

- Trucks
- Buses
- Pickups
- Vans
- Other commercial / fleet vehicles

WHAT MAKES A GOOD TARGET

- Produces, assembles or integrates trucks, buses, pickups, vans or other commercial vehicles
- Connected-vehicle / IoT capability and can work within the required regulatory framework
- Engineering, R&D, ADAS or connected-vehicle teams involved in technology decisions
- A business reason to add advanced driver-safety technology

Competitive Landscape

Three distinct competitor categories — each competing on a different layer.

CATEGORY 01

Local Fleet Tracking

Orange / Metabe3: Vodafone / Arabyati · Telecom Egypt: local GPS providers

THEY COMPETE ON

GPS tracking, fleet management, basic driver behavior. Drowsiness / distraction and driver health monitoring not identified.

D4S OVERLAP

They compete with the tracking layer — not the full D4S concept.

CATEGORY 02

Global Telematics

Samsara · Geotab · Motive · Verizon Connect · Trimble · Webfleet · MiX

THEY COMPETE ON

Telematics, fleet management, driver monitoring, analytics and connected vehicles — mature platforms at scale.

D4S OVERLAP

D4S does not claim to be better overall. Local Egypt adaptation is a potential advantage.

CATEGORY 03

OEM / ADAS Suppliers

Valeo and OEM-embedded ADAS systems

THEY COMPETE ON

AEB, lane assistance, cameras and driver monitoring — already embedded by manufacturers.

D4S OVERLAP

Valeo is an automotive technology supplier, not a fleet-management provider — the closest structural analogue.

COMPETITIVE POSITIONING

TRADITIONAL FLEET TRACKING

"Where is the vehicle?"

A location record — reported after the event

D4S — AI SAFETY LAYER

"How safe is the driver, what risk is developing, and what can we do before a collision?"

THE DIFFERENTIATION IS THE COMBINATION — NOT A FEATURE NOBODY ELSE HAS

Go-To-Market & Pilot

Objective: secure strategic partnerships with manufacturers and integrators — not sell individual tracking devices.

B2B Awareness

LinkedIn content on AI safety & accident prevention, automotive events, exhibitions and technical conferences

Direct Outreach

Engineering, R&D, ADAS, Connected Vehicles & Commercial Vehicle managers — decision-makers, not general contacts

Live Demonstration

Driver monitoring → risk detection → warning → critical event → intervention / alert, with measurable results

Pilot

A small risk-free programme on 3–5 commercial vehicles

Validation

Real-world evidence and case studies built from pilot data

OEM Integration

Move successful pilots into integration agreements

Scale

Use validated results to approach additional OEMs

PILOT SCALE

3–5 Commercial Vehicles

Risk-free proof-of-concept offered to an early OEM or integrator partner.

WHAT THE PILOT MEASURES

Detection Accuracy
Real risk events identified

False Alerts
Find without cause

Driver Response
Digib[illegible]naviour change

Reliability
Uptime in real day cycles

Safety Events
The outcome metric — recorded incidents and near-misses over the pilot period

Pilot → Validation → OEM Integration → Scale

D4S FINANCIAL STRATEGY

Cash Flow Turns Positive in Year 2

A 500,000 EGP prototype investment unlocks operational break-even and rapid cash accumulation.

FUNDING REQUIREMENT

500,000 EGP

Prototype build, initial R&D, and working capital for the first 9 installed units.

Year 1 — LAUNCH	
INSTALLED BASE	9 units
Cash inflows	1,062,000
Cash outflows	(1,033,054)
Net cash flow	+28,946
CASH AFTER DEBT	(20,553.75)

Year 2 — EXPANSION	
INSTALLED BASE	44 units
Cash inflows	4,337,000
Cash outflows	(2,777,429)
Net cash flow	+1,559,571
CASH AFTER DEBT	+1,539,017.50

Year 3 — SCALE	
INSTALLED BASE	204 units
Cash inflows	19,892,000
Cash outflows	(11,121,054)
Net cash flow	+8,770,946
CASH AFTER DEBT	+10,309,963.75

The model crosses into positive cumulative cash during Year 2.

10.3M EGP

YEAR 3 CUMULATIVE CASH AFTER DEBT

Business & Revenue Model

D4S becomes a **Technology Supplier / OEM Technology Partner** – not a direct-to-fleet product.

OLD MODEL D4S → Fleet / Transportation Company → Truck

NEW MODEL – B2B2B

01 D4S
AI safety technology supplier

→

02 OEM / Authorized Integrator
Primary customer or channel partner

→

03 Commercial Vehicle
System integrated into the vehicle

→

04 Fleet
End user / beneficiary

→

05 Driver
Protected

SUPPORTED REVENUE STRUCTURES

Hardware + Annual Software License
Upfront unit sale plus a recurring annual license per vehicle

Integration Fee + Per-Vehicle SaaS
Integration charged per vehicle, then recurring software per unit

OEM Partnership / Revenue Sharing
Manufacturer embeds D4S and shares programme revenue

REGULATORY POSITION
Deployed through licensed manufacturers and authorized integrators, with compliance handled as part of vehicle integration. Exact requirements still need verification.

95,000
Selling price / vehicle (EGP)

50,000
Upfront COGS – hardware 42,000 + integration 8,000

23,000
Annual SaaS / vehicle – recurring software fee

45,000
Upfront gross margin per vehicle

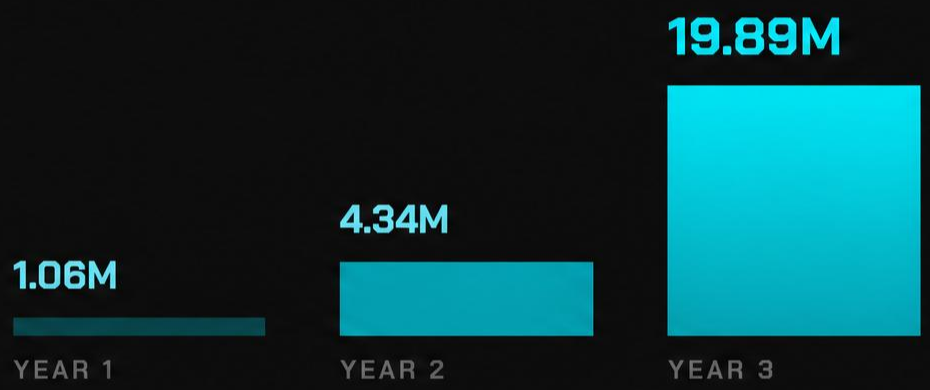
Financial Model

3-year pro forma, EGP · Annual fixed OpEx **574,650** · Corporate tax **22.5%**

INSTALLED VEHICLES (YEAR-END ACTIVE UNITS)



TOTAL REVENUE (EGP)



FINANCIAL METRIC (EGP)	PROTOTYPE	YEAR 1	YEAR 2	YEAR 3
Total Revenue	—	1,062,000	4,337,000	**19,892,000**
Gross Profit	—	612,000	2,587,000	**11,892,000**
Net Profit	—	28,946	1,559,571	**8,770,946**
Cumulative Cash After Debt	(49,500)	(20,554)	1,539,018	**10,309,964**

SOURCE: D4S FINANCIAL PLAN · PROTOTYPE COST / INITIAL DEBT 49,500 EGP

Why D4S?

Aligned with UN Sustainable Development Goals

3 **Good Health & Well-being**

Reducing accidents means fewer injuries and deaths. Real-time monitoring helps keep drivers safer.

11 **Sustainable Cities & Communities**

Fewer accidents mean safer roads and less crash-related traffic, improving urban mobility.

8 **Decent Work & Economic Growth**

Better driving behavior reduces accidents, maintenance costs, and delays—creating a safer work environment.

13 **Climate Action**

Improved driving behavior can reduce fuel consumption by 10–15%, lowering carbon emissions.

9 **Industry, Innovation & Infrastructure**

Driver monitoring and data analysis modernize the transportation sector.

16 **Peace, Justice & Strong Institutions**

Real-data driver scoring replaces subjective judgment, making evaluation fair and transparent.

VISION

From simply tracking vehicles

to predicting and preventing accidents.

D4S — An OEM-integrated AI safety technology for commercial vehicles.

Instead of asking individual fleet operators to deploy a regulated vehicle-control system, D4S partners with licensed automotive manufacturers and authorized integrators to embed the technology into commercial vehicles.

D4S

DRIVE FOR SAFETY

Assiut Motorsport Team
Assiut University - Egypt